

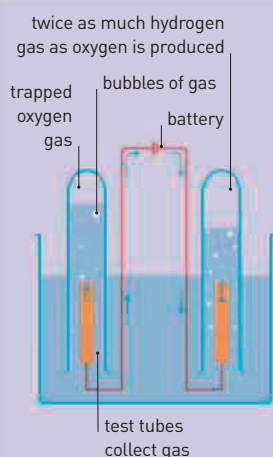
“WE FOUND THE VERY SUPERIOR SPECIES OF THE VESPERTILIO-HOMO... THEY... APPEARED IN OUR EYES SCARCELY LESS LOVELY THAN THE REPRESENTATIONS OF ANGELS BY... PAINTERS.”

A description of the inhabitants of the moon in the *New York Sun*, 1835



The Great Moon Hoax in the *New York Sun* caused a sensation and fooled many with illustrations of people on the Moon.

If it had been completed it would have been a genuine forerunner to the computer (see pp.184–85), because it would have been programmable, had a memory, and could have been programmed to perform many tasks besides simple calculation.



ELECTROLYSIS OF WATER

Electrolysis splits water into its component elements (hydrogen and oxygen) by passing an electric current through it. Hydrogen gas gathers in the test tube above the negatively charged electrode and oxygen gas gathers in the test tube above the positively charged electrode. Twice as much hydrogen is released.

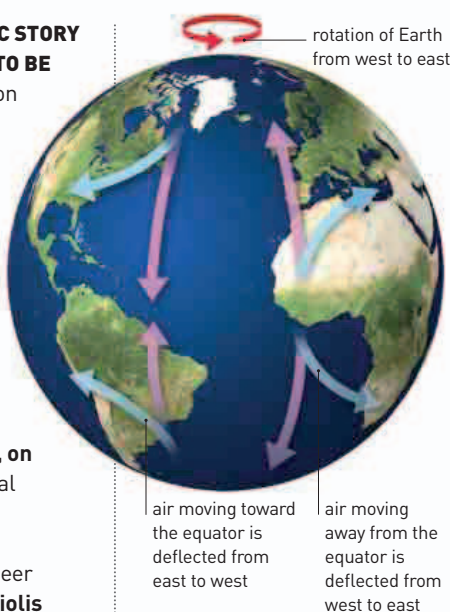
A POPULAR SCIENTIFIC STORY OF 1835 TURNED OUT TO BE

A HOAX. The Great Moon

Hoax was a series of stories in the *New York Sun* newspaper that covered the “discoveries” of the famous British astronomer **John Herschel** (1792–1871). The illustrated articles claimed there was **life, and even civilizations, on the Moon**. It was several weeks before the joke was revealed.

In 1835, French engineer **Gaspard-Gustave Coriolis** (1792–1843) identified what is now known as the **Coriolis effect**—the effect of Earth’s rotation on both wind and water moving across Earth’s surface. Winds do not follow a straight path, but veer to the east in the Northern Hemisphere and to the west in the Southern Hemisphere, sometimes curling around in loops in clockwise and counterclockwise circulations. The same is true of ocean currents.

Knowledge of the history of Earth took a step forward when **two geological periods were identified** (see 1821–22). English geologist **Adam Sedgwick** (1785–1873) proposed the



Coriolis effect

Earth’s rotation is the cause of the Coriolis effect. One of the main impacts of it is the deflection of winds and currents in the ocean.

Cambrian period, naming it after Cambria, the Latin name for Wales, where Britain’s Cambrian rocks are best exposed. Scottish geologist, Roderick Murchison (1792–1871), proposed the **Silurian period** in the same year, naming it after an ancient Celtic tribe, the Silures, and the two

Talbot exposes photo negatives

Fox Talbot’s calotype process recorded a negative image, from which a large number of positive prints could be made.

Darwin’s notebook

Darwin recorded his observations on the Beagle voyage (1831–36) in notebooks. His records of this trip helped him develop his theory of evolution over the next 20 years.

presented their findings the same year in a joint paper.

On September 16, 1835, **Charles Darwin** (1809–82) landed on the **Galapagos Islands** for the first time.

His visit to the islands had a significant impact on his theory of evolution (see 1857–58).

In England, inventor **Henry Fox Talbot** (1800–77) produced the **world’s first photographic negative** (dark and light reversed). Although French physicist Louis Daguerre (1787–1851) went public with his process known as daguerreotype first (see 1837), this process produced a one-off positive photograph. Talbot’s process, called the **calotype**, produced a negative photograph



from which many positive prints could be made.

On the other side of the Atlantic, in 1836, American inventor **Samuel Morse** (1791–1872) developed a code (later called **Morse code**) for sending messages by telegraph, with each letter of the alphabet represented by a unique combination of short pulses (dots) and long pulses (dashes).

Another American, **Samuel Colt** (1814–62), was granted a US patent for the revolver. This gun could fire six shots in quick succession by using a revolving cylinder that automatically moved a new bullet into the firing position after each shot.

Charles Babbage begins work on his Analytical Engine

1835 Gaspard-Gustave Coriolis identifies what becomes known as the Coriolis effect

August 25, 1835 The Great Moon Hoax is published in the New York Sun newspaper

September 16, 1835 British naturalist Charles Darwin arrives at the Galapagos Islands

1835 British geologist Adam Sedgwick proposes the Cambrian period

1836 German physiologist Theodor Schwann identifies the first animal enzyme—pepsin

1836 Samuel Morse develops Morse code

German geologist Friedrich Alberti identifies the Triassic period

1835 British astronomers Caroline Herschel and Mary Somerville become the first female members of the Royal Astronomical Society

1835 Henry Fox Talbot produces the world’s first photographic negative

1835 Roderick Murchison proposes the Silurian period

1836 Irish chemist Edmund Davy discovers acetylene

1836 Samuel Colt is granted a US patent for the revolver